Namit Bhutani

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Research Interests

Interactive graphics bringing rendering, simulation and modelling together through real-time and spectral rendering, inverse procedural modelling of captured scenes, and physics-based simulation, with GPU algorithms and ML.

Education

Birla Institute of Technology and Science (BITS) Pilani, Hyderabad Campus 2022 – 2027 (expected)
B.E. (Hons.) Computer Science and M.Sc. (Hons.) Economics, dual degree. Advisor: Tathagata Ray CGPA 8.92 / 10
Relevant coursework: Computer Graphics, Machine Learning, Algorithms, Probability and Statistics, Operating Systems

Publications

Real-Time Spectral Photoelasticity via Stochastic Hero-Wavelength Ray Marching and Stress-Aware Temporal Reprojection
N. Bhutani, T. Ray, S. Pattanaik
Pacific Graphics Conference Papers (Eurographics). **Oral talk and poster**. [paper] [project] [code] 2026

Generative Discourse Parsing for Indian Law
C. R. Chaitra, **N. Bhutani**, S. Kulkarni, S. R. A. V. Sagi, S. Pandey, R. Yalavarthy, D. Chakraborty, P. Upadhyay
Artificial Intelligence and Law (Springer). [paper] 2026

Research Experience

Dynamic Graphics Project, University of Toronto June 2026 – Present
Research Intern and BITS undergraduate thesis. Advisors: Karan Singh; Tathagata Ray (BITS) Toronto, Canada

- *Structural editing of captured scenes* (targeting SIGGRAPH 2027). Designed and built a library of structure models (repeated sections, chains, rods) and relations that measure their parameters from a Gaussian-splat capture, so edits repeat captured units instead of stretching them; geometric gates judge each edit and an optimiser refines appearance and hole fills.
- Testing where a language-model agent helps most (editing front end, segmentation) so the tool needs minimal user input.
- *Interactive splat tools.* Houdini plugin to paint and select 100K+ splats at 60 FPS, with DINOv3 features for segmentation.

BITS Pilani and University of Central Florida August 2025 – May 2026
Undergraduate Researcher. Advisors: Tathagata Ray, Sumanta Pattanaik

- *Real-time spectral photoelasticity.* Adapted hero-wavelength sampling to stress-induced birefringence and designed a stress-aware temporal filter that removes ghosting under moving loads. Runs at 120+ FPS on a laptop GPU, 8.55× faster than a sixteen-wavelength baseline, validated against a real polariscope photograph. First-author paper and oral talk at Pacific Graphics 2026.

BITS Pilani August 2024 – July 2026
Undergraduate Researcher. Advisors: Dipanjan Chakraborty, Prajna Upadhyay Hyderabad, India

- *Generative discourse parsing for Indian law (LegalTree).* Built the group's demo comparing a fine-tuned T5 parser with GPT-4 and GPT-4o prompting (interactive trees, Graphene triplet extraction, retrieval QA over 1,000+ documents); for the journal revision, built the delinearization evaluation (BLEU-4, BERTScore, ROUGE).

Open Source

Google Summer of Code, Blender Foundation June 2025 – September 2025
Contributor. Mentor: Hans Goudey

- *Sculpt-mode performance on million-vertex meshes.* Reordered mesh data for cache locality and cached its BVH (85% faster sculpt-mode entry), and ZSTD-compressed undo data (at least 2× smaller). Shipped in Blender 5.0 after core-developer review.

Teaching and Awards

Teaching Assistant, BITS Pilani: BITS F102 Innovation and Design Thinking; BITS F111 Thermodynamics

GCC Travel Scholarship, BITS Pilani (INR 23,867) to present at Pacific Graphics 2026

Industry Experience

AlphaSense, AI Engineer Intern (incoming), Bangalore January 2027 – June 2027

Vidcentum Technologies, Software Engineering Intern, Hyderabad May 2024 – July 2024

Selected Projects

procLan: GPU terrain generator that meshes a domain-warped 3D noise field with Marching Cubes in compute shaders, so caves and overhangs, which heightmaps cannot represent, are shaped live. 100K+ polygons at 60+ FPS. [code] 2025

Gaussian splats in Unreal Engine: C++ plugin that brings Gaussian-splat captures into Unreal's Niagara, with a PLY loader, a custom data interface and a render-thread proxy that uploads splats to GPU buffers. Built for Prof. Karan Singh. [code] 2026